Dundalk Institute of Technology

Interim Report on Investigations on Multi Disease Prediction in Dairy Cattle Using Machine Learning on French Dairy Cattle Dataset

Student :
Vishnu Bhuvanachandran Nair Rajeswari Amma
No. D00275640

Supervisor:
Dr. Zohaib Ijaz

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Contents

1 Background **2**
1.1 Objectives of the Study 3
1.2 Strengths of This Study 3
1.3 Limitations and Weaknesses 3

2 Literature Review **4**
2.1 Challenges in Previous Studies 6

3 Exploration of data and Methods **6**
3.1 The Dataset 6
3.2 Artefact and Methodology 8

4 Proposed Future Analysis **10**

1 Background

Development of agricultural sector is an integral part of every nations economical development which directly impacts GDP. Apart from the farming and crop development, livestock farming forms the inevitable part of agricultural economies in European Union, of which the 1.4% of total GDP was contributed by agriculture [1] . Milk production from dairy cattle accounts to 14% of the total agricultural production after fruits and vegetables [2]. In this regard, the health and well-being of cattle directly impacts not only the the global economic development but also the food security and the animal welfare. This in turn brings out the significance of health care of animals, which can be effectively solved by proper earlier systematic disease diagnosing systems.

The timely disease diagnosis via. conventional methods such as veterinarian intervention has many challenges. One such major challenge is the location of the cattle farms, majority of them are in rural areas where the access to the medical facility and veterinary specialists are limited. The inability of the farmers to detect the disease symptoms due to the lack of knowledge and awareness adds to the other challenge associated with the timely diagnosis of diseases such as mastitis, oestrus, lameness, and calving [3] [4]. This in turn results in the preventable disease conditions progressing to severe sufferings for the cattle and which may result in mortality resulting in huge economic losses.

One of the major achievements in the field of Machine Learning is the advancement in the area of disease predictions. These predictions are carried out with respect to medical images, sensory data, or federated data. Among these medical predictions, the disease prediction of animals has gained attention in the recent past. This is due to the fact that there are many regions like Austria, Switzerland and Ireland, where the majority of investments lie in agriculture and farming, and where medical diagnosis of cattle is a serious problem. This is due to various factors such as (1) availability of medical facilities, (2) remote location of farms, and (3) awareness of farmers. Therefore, it is inevitable for the agriculture industry to have a prediction system that can effectively classify and predict cattle diseases.

Therefore this work aims to address these challenges by developing an data driven machine learning algorithm for disease prediction using behavioral and activity-level data collected from cattle. This prediction is carried out by multi-output classification, which allows simultaneous prediction of multiple diseases the same time with respect to the behavioral data of the cattle. The outcome of this approach which is the early prediction of the disease can effectively offer a scalable solution for early warning enabling farmers and veterinarians to intervene promptly and reduce the severity and spread of diseases.

The main focus of this current study is to predict four key diseases—mastitis, oestrus, lameness, and calving using the structured behavioral data collected from the french diary dataset. In order to address the data imbalance and feature correlation, oversampling methods such as SMOTE will be checked. The modeling starts with the traditional models like Random Forest and progresses towards the deep learning architectures such as MLP, aiming to optimize both accuracy and interpretability. The effectiveness of time series approach also will be checked in the current work to analyze the effect of temporal patterns involved using method such as LSTM.

1.1 Objectives of the Study

The main objective of the study is to develop a multiple output disease prediction model using machine learning algorithm from the available merged datasets of cattle from the French agricultural database. The specific objectives are as follows:

- Select the ample amount of data and carry out the data cleaning and misisng data handling.
- Carry out EDA for the data to analyze the trends of the data.
- Subject the data to conventional models such as random Forest, Gradient Boosting, Gradient descent, and deep learning methods such as MLP, Tab net.
- Check the possibility of time series analysis via. neural networks such as LSTM to understand the importance of underlying temporal pattern.
- Investigate the effect of oversampling method such as SMOTE which could compensate for the imbalance in the dataset.
- Deployment of the model and artefact building.

1.2 Strengths of This Study

- Larger dataset (300+ cows) with behavioral data with time characteristics
- Multi-output prediction, enabling simultaneous detection of multiple health events at the same time compared toindividual training performed by previous studies.
- Data imbalance handling via. SMOTE (Synthetic Minority Oversampling Technique) integration and their effectiveness checking in the disease prediction cases.
- Comparison between traditional models and deep learning models such as MLP (Multi Layer Perceptron) and TabNet
- Analysis of the effectivenes of the time dependance via. the neural methods like LSTM (Long Short Term Memory)

1.3 Limitations and Weaknesses

- The data imbalance due to high number of no-disease condition compared to sick condition.
- Impossibility associated with the validation of the results with the real time cases.
- Missing value handling in the dataset can cause discontinuity in the time series affecting the temporal pattern.

2 Literature Review

The researchers have explored various models for disease predictions with the help of real time sensory data, federated data and manually collected data in the recent past. This section reviews the key contributions in the research investigations regarding the multiple disease predictions, models used and the methods to handle the imbalance in the data.

One of the successful algorithms developed in the recent past was deployed in the Russian Scientific Agro Engineering Centre, which detects symptoms leading to many diseases such as nervous, infectious diseases, and so on [5]. The studies reported by investigators faced many issues such as: (1) shortage of data, (2) biased data, (3) generalizability of breeds, and so on. Prior to the above study, Rober et al., 2020 developed a model for predicting mastitis infection patterns in dairy cattle using the SMOTE (Synthetic Minoprity OverSampling Technique) technique to develop synthetic data to adjust for imbalanced data [6]. Even though the bias in the minority cases was handled, the SMOTE technique did not significantly improve the model.

Subsequently, Heidi et al., 2020 approached the problem with an ensemble learning method to predict dairy cattle survival and found that regression ensemble performed better compared to others in the ensemble method [7]. This study was one of its kind where the regression ensemble predicted with the highest precision with AUC (Are Under Curve)ranging from 0.561 to 0.731, followed by the naive ensemble method.

Following this, adapting the Apriori algorithm was another development introduced by Chandana et al., 2020 to help veterinarians diagnose infectious diseases in cattle from electronic records [8]. It was found that the algorithm performed better compared to SVM. However, as usual, imbalance, lower data quantity, and generalizability remained as drawbacks.

Apart from the above studies, a new algorithm was developed for a dataset involving not only cattle features but also environmental features [5]. In addition, prediction of behavioural issues in Austrian dairy cows under subacute ruminal acidosis was attempted by Wagner et al., 2020 [9]. In this study, characteristics of 28 cows were monitored for 9 weeks, and it was found that KNN gave the highest accuracy for detection. However, lesser data content was a drawback.

Following this, with the same dataset, Wagner et al. investigated the reproductive diagnosis of cattle with respect to the daily activity of cows using BOSS (Bag of Symbolic Fourier Approximation Symbols) and FBAT (Fourier Bases Approximation with Thresholding)techniques [10]. BOSS resulted in high speed, and FBAT gave high accuracy. However, generalizability and false alarm presence were the disadvantages of the study.

Furthermore, to avoid limitations in data quantity and quality, a foundation dataset was developed by Lardy et al., 2022 comprising more than 300 cattle [11]. This foundation dataset contains activity patterns and health patterns and was cleaned, curated, and ensured to be viable for machine learning studies. In this case, the absence of features such as environmental data and 17.3 percent missing data were the drawbacks.

Finally, the investigation by Lardy et al., 2023 utilized these 5 datasets together to classify diseases such as lameness, mastitis, acidosis, accident, oestrus, calving, and so on [12]. The classification task aimed to predict the event rather than the disease and found that reproductive events had the highest prediction accuracy accounting for 81% using a Random Forest classifier compared to other classifiers.

In addition to the above disease prediction practices concerning cattle, the methods through which multiple predictions were carried out require an extensive review to understand potential possibilities. Various investigations reported models on multiple predictions via simultaneous multi-output predictions and independent multi-output prediction. The former focuses on predicting multiple outcomes together, and the latter on predicting multiple outcomes independently.

Multilabel chronic disease prediction is one such study reported by Zhang et al., 2019 for the prediction of chronic diseases [13]. GroupNet, GoogLeNet, LSTM (Lomg Short Term Memory), SVM (Support Vector Machine), and Decision Trees were involved in the comparative analysis for the prediction of 110,300 records with 62 features. Even though GroupNet outperformed other models, the associated computational cost and imbalance of the data were drawbacks.

Joint classification and regression using DM2L (Deep Multi task Multi channel Learning) is another approach adopted by Liu et al., 2019 for predicting Alzheimer's disease [14]. The approach analyzed MRI (Magnetic Resonance Imaging) images and demographic information of patients, including age, gender, and education. Even though the model was efficient, the computational cost was high, and important features such as genetic information, which could improve the accuracy, were absent.

Later in 2021, another multilabel classification for medical purposes was attempted by Liu et al., 2021, for predicting protein subcellular localization [15]. The significant difference in this model compared to other investigations was the incorporation of different feature extraction and dimensionality reduction techniques, which made it more advanced.

Apart from disease prediction studies, multi label prediction is also employed in NLP (Natural Language Processing) for classifying emotions from texts [16]. Multi label KNN (K Nearest Neighbor) is the method used here, which further divides the operations using sample- and label-based MLKNN (Multi Label K Nearest Neighbor) for grammar features in adjacent sentences and the label correlation, respectively. The main drawback of the model is the influence of label correlation on computational cost and lower performance associated with small texts with single labels.

Subsequently, a hybrid multilabel model was introduced by Priyadarshini et al., 2023, with the main aim of compensating for the imbalance contributed by the minority data in datasets [17]. This involved synthetic data incorporation by adaptive synthetic data and ensemble learning. Despite the fact that data imbalance was compensated, the scalability and generalizability of the synthetic data and computational cost added to the cons of this technique.

Apart from the above studies, the multi-classification study by Gopisetti et al., 2023 predicted the presence of various diseases based on patient data using various algorithms [18]. The investigation achieved approximately 99% and 98% accuracy for chronic kidney and diabetes diseases respectively. The former was achieved with the Random Forest, while the latter was achieved using the Decision Tree. Even though the model had high accuracy and potential deployment with Streamlit, information on data imbalance, normalization, feature engineering, and generalizability was lacking.

Furthermore, multi-output and class classification using Gaussian processes was adopted by Chunchao et al., 2023 using a novel approach with linear model coregionalization and Gumbel noise [19]. This advanced method classified each output as a class and incorporated sub-sampling to reduce computational costs. Unlike other models, high scalability is the major highlight of this process. However, feature loss during image downscaling

and non-viability in multilabel classification are the cons of this work.

This shortcoming was further investigated by Rahman et al., 2024 in his investigations on breast cancer prediction using machine learning with the help of a multi-output classifier [20]. The neural networks showed highly accurate predictions in surgery and chemotherapy and very low performance for radiotherapy treatment. This was due to the varying complex parameters involved in the radiotherapy treatments.Data bias and the lower accuracy in radiotherapy due to varying treatment protocols adds to the drawback of the investigation.

As different from the above studies, multiple disease prediction in telemedicine is carried out using LSTM by Divya and Jeba 2024 [21]. Two layers of 64 and 128 units in LSTM were applied to study the prediction and it was observed that the underlying patterns pertasining to the cause of disease were effectively understood by the model resulting in an accuracy of 98.51%. Very low loss value of 0.0842 ads to the highlight of the investigation. The generalizability and computational cost were the disadvantages associated with the process.

2.1 Challenges in Previous Studies

- Data imbalance: Majority of the disease shows (0) expressing the healthy condition
- Lack of generalizability: Majority of the studies done on specific breeds which is in shorted numbers failed to generalize with other breeds.
- Size of the dataset size: the lower size of the dataset not only reduces the generalizability , but also the scalability and accuracy of predicition.
- Time dependent analysis: The underlying temporal patterns brought out by the cattle with respect tot eh time are least explored.
- Feature correlations: Correlation between teh features are least explored in the previous studies.
- Predictions: The multi output predicitions using deep learning methods are least explored which may capture the underlying complex patterns resulting in accurate analysis.
- Oversampling: the effect of oversampling in the disease prediction by handling the imbalance is least explored.

3 Exploration of data and Methods

3.1 The Dataset

The dataset for the analysis was taken from the source published by the National Research Institute for Agriculture, food and Environment of France. The dataset is available to public for analyzed, which is sensor based real data collected from 386 dairy cows and the same is anonymized with numbers [12]. The dataset contains the features listed below in the table 1.

Features	**Particulars**	**Measurement Units**
Cow	Cow identification number	Identification number
Date	Sensor reading date	Date-time
Hour	Observed hour	Hour
In Alleys	Time in seconds spent in alleys	Time in seconds
Rest	Time in seconds took for resting	Time in seconds
Eat	Time in seconds took for eating	Time in seconds
Activity Level	Activity level calculated from alleys, rest and eating behaviour	Simple Integer
Oestrus	Disease condition	Binary (0 and 1)
Calving	Disease condition	Binary (0 and 1)
Lameness	Disease condition	Binary (0 and 1)
Masitits	Disease condition	Binary (0 and 1)
Disturbance	Disturbance observed	Binary (0 and 1)
Accidents	Accidents observed	Binary (0 and 1)
Management	Dairy management change observed	Binary (0 and 1)
OK	Overall health observed	Binary (0 and 1)
LPS	Lipopolysaccharide inflammation observed	Binary (0 and 1)
Acidosis	Acidosis observed	Binary (0 and 1)
Other Diseases	Other diseases observed	Binary (0 and 1)
Mixing	Cow herd mixing observed	Binary (0 and 1)

Table 1: Dataset Features

The dataset was merged from the 4 datasets available from the public repository which contains 2,351,345 rows. The features such as LPS and acidosis are eliminated due to the presence of Nan values accounting to more than 90 percentage. In addition to this, the features management and OK are also eliminated as it is not relevant to the current study. Even though an ample quantity of data is present, the condition having presence of disease is very low compared to non-disease state which can contribute to imbalance in the data. The count of these conditions, which are the presence of disease indicated by 1 and healthy condition represented by 0 is portrayed in table 2.

Disease Condition	**Absent**	**Present**
Oestrus	2,343,401	7,944
Calving	2,347,049	4,296
Lameness	2,348,129	3,216
Mastitis	2,350,289	1,056
Other Disease	2,349,329	2,016
Accidents	2,350,985	360
Disturbance	2,037,783	313,562
Mixing	2,349,617	1,728

Table 2: Count of Disease Conditions

At current stage, the dataset is preserved despite the fact that there are missing values in the dataset. This was due to the fact that the in depth research on effective incorporation of missing data is in progress.

3.2 Artefact and Methodology

The artefact and the methodology for the project is presented in the figure 1. The artefact flow is maintained as per the instructions and guidelines provided by the Institute. The ethical approval was taken from the Institute via. proper channels. The repository for data storage was created in the Institute Gitlab as per the guidelines of the Institute. In addition to the data regarding the project, the meeting minutes and the review documents were also uploaded to the repository and the markings of the same is shown in figure 1 below.

The project commenced with the data collection and baseline study via. literature review. The data was collected and merged from the publicly available source described in the section 3.1. This was then subjected to cleaning operation. The data was understood further by analysing the missing data and disease presence and absence proportions. This step was very important to understand the imbalance in data. This will further be subjected to missing value handling in order to impute data or eliminate the unwanted data depending on the nature of missing data.

Subsequently, this data will be subjected feature engineering and then to exploratory data analysis(EDA) to understand the trend of data. The main problem associated with the cattle disease prediction data is the imbalance in the disease conditions which can lead to unstable predictions. This was compensated with the synthetic data addition using SMOTE technique. The conventional dataset and the newly generated dataset will be subjected to Machine Learning Algorithms.

The baseline algorithms such as Random Forest, Gradient Boosting etc., will be applied and compared at the initial level for multi output prediction of diseases such as lameness, oestrus, calving and mastitis. This in turn will be subjected to deep learning models such as MLP and TabNet which could effectively bring up the underlying patterns in the data. The non linear pattern recognition using the neurons is the another added advantage of these models. In addition to the above, the usefulness of time dependent analysis also will be checked using neural network LSTM.

The best method with optimum evaluation metric will be selected and also the effectiveness of SMOTE in disease prediction dataset also will be evaluated and compared. The selected model will be deployed successfully.

The detailed work plan of the above methodology is presented in the figure 2.

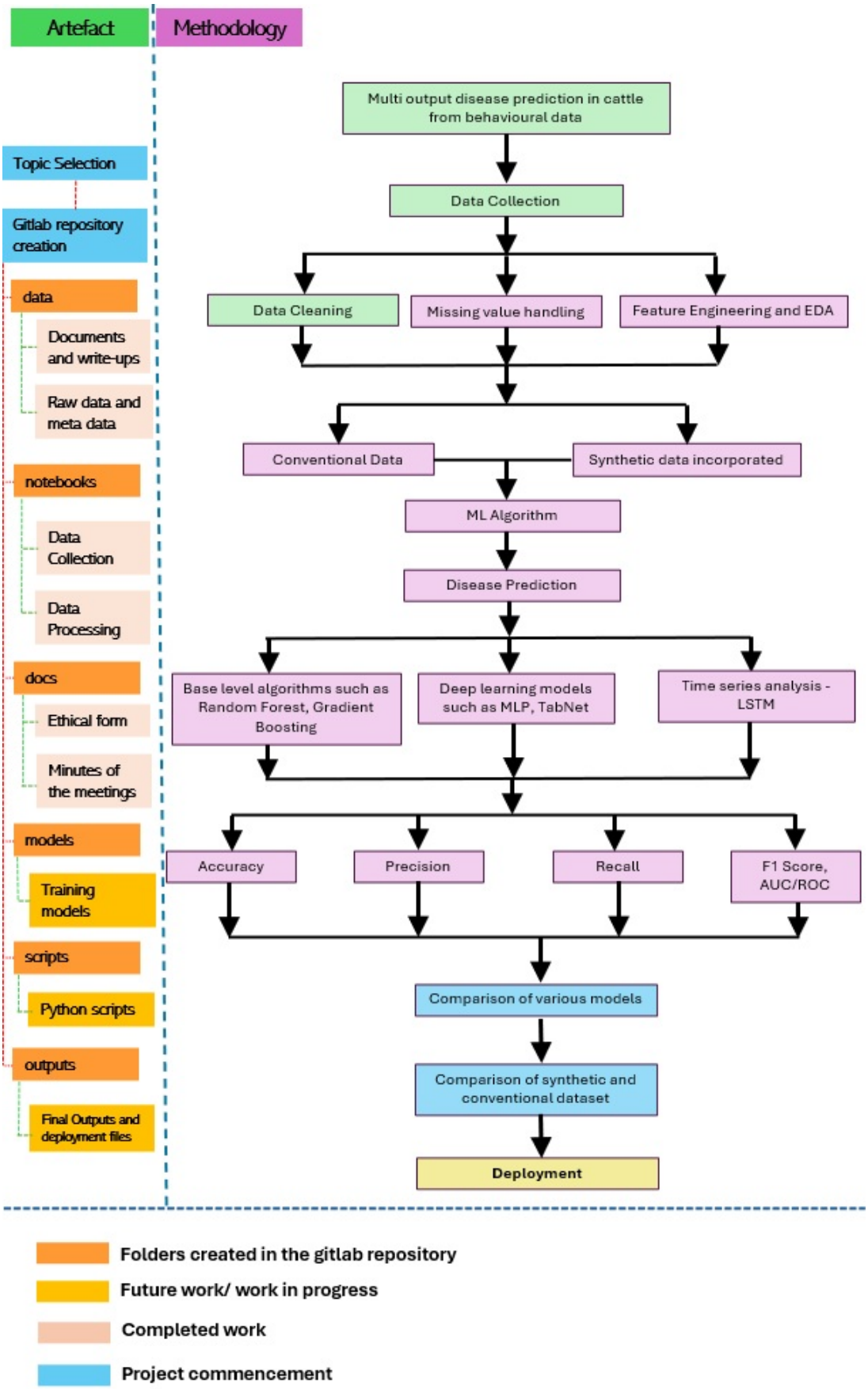

Figure 1: Artefact and Workflow for Cattle Disease Prediction Methodology

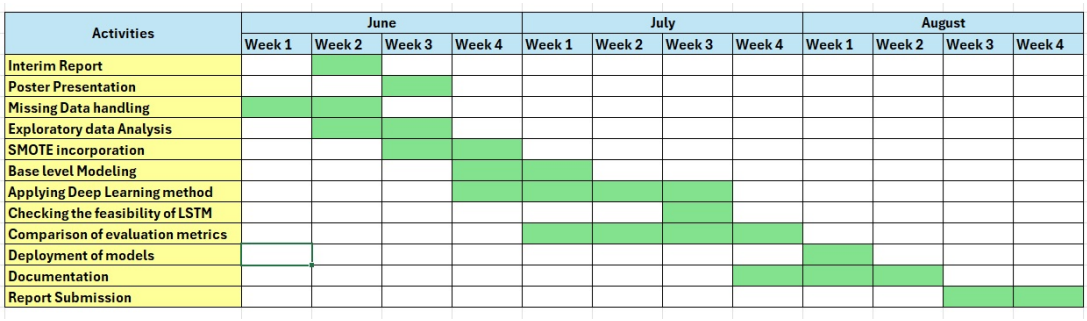

Figure 2: Workplan

4 Proposed Future Analysis

The derailed work plan for the project progress is provided in the figure 2. The future work will progress in such a way that the effective handling of missing data via. imputing or deletion with respect to the missing data. Consequently, the data imbalance will be compensated by SMOTE technique. The conventional data and the newly generated data will be modeled with the baseline models. Following to this, the data will be modeled using beep learning models. In addition to the above, the data will be checked for the time dependent analysis using neural networks such as LSTM. The comparison to find the best multi output regressor falls in the next step of the process. The successful deployment of the model will be the final step of the investigation.

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